

ITT320 INTRO TO COMPUTER SECURITY

Chapter 11 : Legal & Ethical Aspects

Universiti Teknologi MARA



Objectives

- Cybercrime and Computer Crime
- Intellectual Property
- Privacy
- Ethical Issues
- Ethical Hacking & Penetration Testing
- Legal Aspect



Introduction to Legal & Ethical Aspect

- Many types of computer attacks can be considered crimes and, as such, carry criminal sanctions .
- After studying this chapter, you should be able to :
  - Discuss the different types of computer crime
  - Discuss the different types of computer crime
  - an overview of key issues in the area of privacy
  - Legal Aspect and Act governed by the law enforcement
  - Compare and contrast various approaches to codifying computer ethics and Penetration Testing

Types of Computer Crime

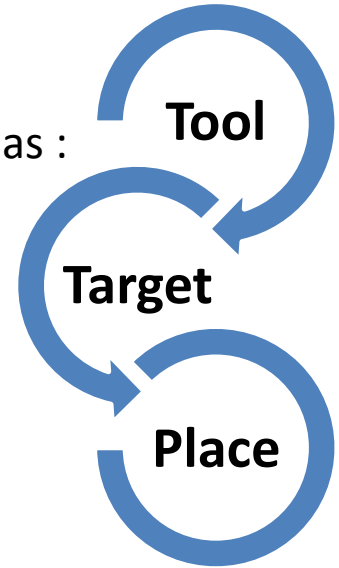
- Computer Crime or Cyber Crime is a criminal activity in which computers or computer networks are a tool, a target, or a place of criminal activity.

- The terms connotations:

Cybercrime - The use of networks specifically to Commit the crime

Computer crime - May or may not involve networks

Computer as :





Types of Computer Crime

Categories of computer crime is based on the role that the computer plays in the criminal activity :

1. **Computers as targets** – Targets a computer system, to acquired information stored, to control the target system, to alter the integrity of data
2. **Computers as storage devices** – Computer can be used as a passive storage medium such as to store stolen password, credit card and pirated commercial software
3. **Computers as communications tools** – Illegal sales of prescription drugs, alcohol, guns, fraud, gambling etc.

Many crimes will involve perpetrators who are remote from the target system.

Cyber Crime

Types of Cybercrimes based on International Convention on Cybercrime:

1. ***Illegal Access*** – access to computer system without right
2. ***Illegal Interception*** – interception w/o rights, towards computer data
3. ***Data Interference*** – damaging, deletion, alteration, suppression of computer data
4. ***System Interference*** – serious hindering w/o right
5. ***Misues of Devices*** – used for the purpose of committing any of the offences from 1 - 4



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Cyber Crime

Types of Cybercrimes based on International Convention on Cybercrime:

1. ***Computer Related Forgery*** – alteration computer data resulting inauthentic
2. ***Computer Related Fraud*** – Causing a loss of property to another person
3. Offenses related to child pornography
4. Infringement of Copyright and related rights
5. Attempting and aiding or abetting



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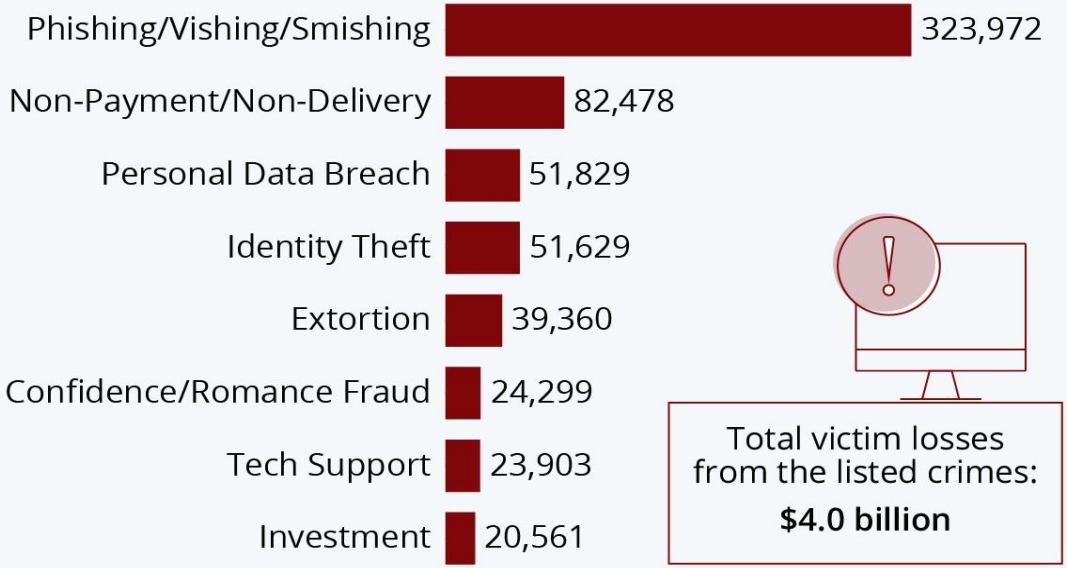
Ransomware (securityintelligence.com)

Most common types of Hacks and Malware:

1. Phishing Attacks
2. Vishing Attacks *Phishing attacks by phone*
3. DDoS Attacks
4. Identity Theft
5. Ransomeware Attacks

The Most Common Types of Cyber Crime

Number of Americans who fell victim to the following types of internet crime in 2021



Source: The FBI's Internet Crime Complaint Center

cc

statista

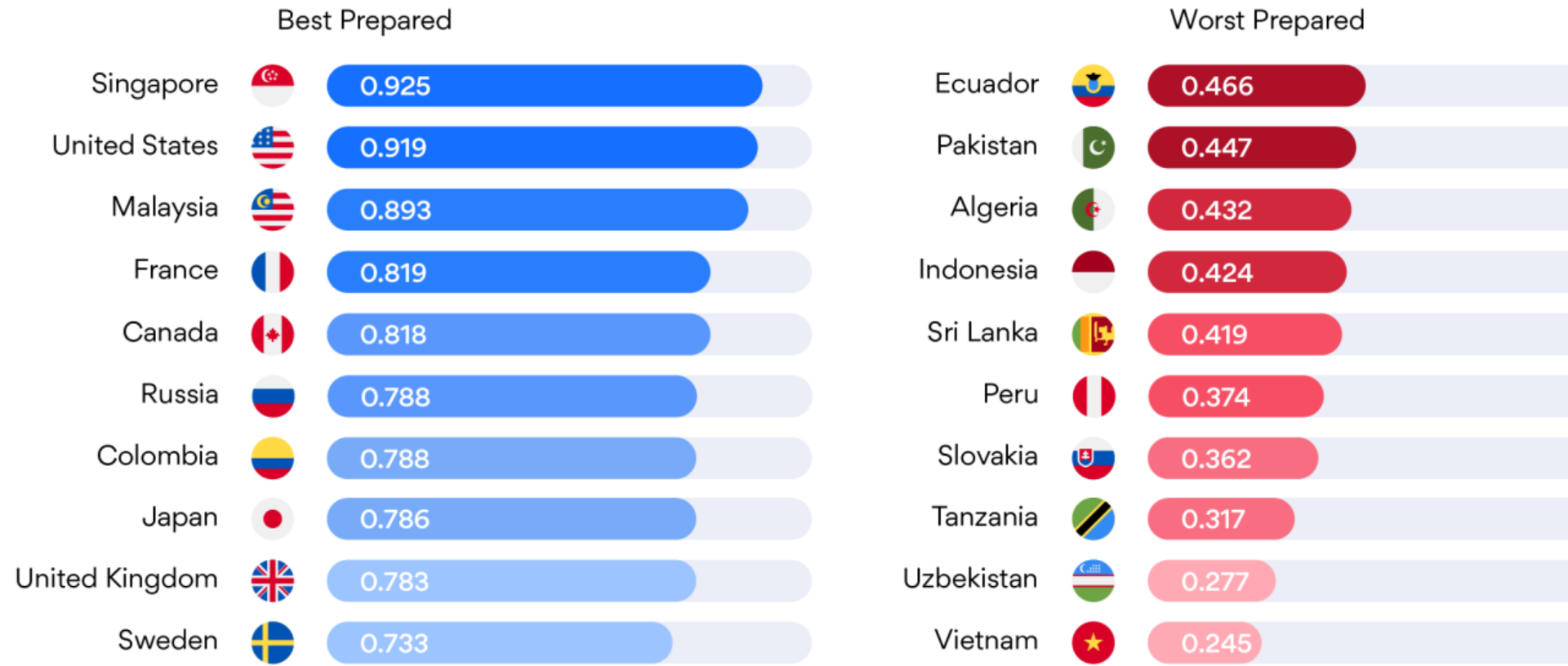
Cybercrime in Malaysia

79% of Malaysian organisations were targeted by ransomware in 2021, with 64% of attacks resulting in the encryption of data.

- ✓ 20K Cybercrimes reported in 2021
- ✓ 2017-2021 Total Amount Lost is RM2.23 Billion
- ✓ January to July 2022 – 11,367 reported Cases



10 Best and Worst Prepared Countries for Cybercrime



https://www.pcmag.com/news/which-countries-are-best-prepared-for-cybercrime-response

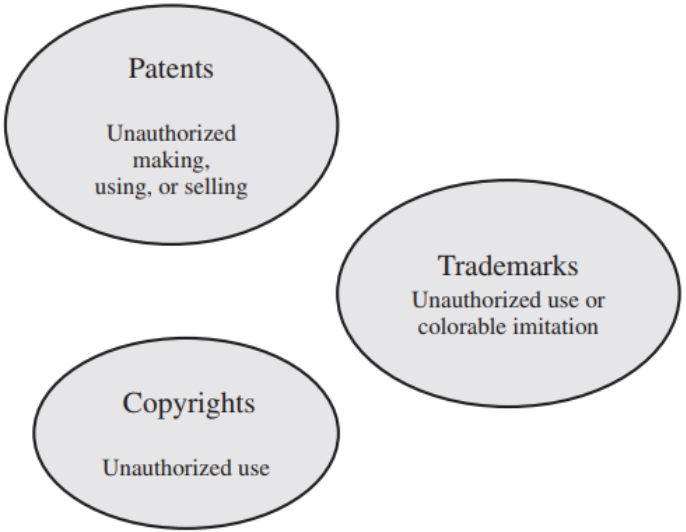
Intellectual Property

There are Three types of Intellectual Property in regards to computer security :

Copyrights | Trademarks | Patents

The legal protection is against Infrigement

(the action of breaking the terms of a law)



Intellectual Property – Copyrights

Copyrights

- ✓ Protects the tangible or fixed expression of an idea
- ✓ **Example** – Musical works, graphic, audiovisual, sound recordings
- ✓ **Software-related works** – Computer software, software documentation,

Manual, training manuals

Protected Against Infringement :

1. Reproduction Rights
2. Modification Right – Derivative-works right
3. Distribution Right – Owner can sell, rent, lease, lend copies
4. Public-performance right
5. Public Display right – film, slide, television image



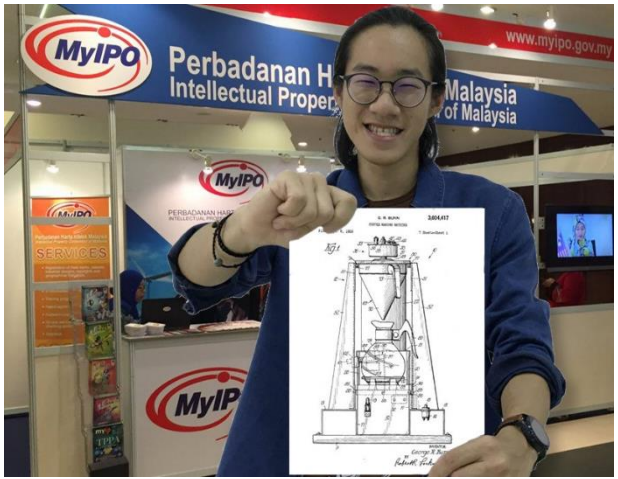
Intellectual Property – Patents

Patents

A patent for an invention is the grant of a property right to the inventor.

Three types of Patents:

1. **Utility Patents** – To anyone who invents useful process, machine, article of manufacture, useful improvement
1. **Design Patents** – To anyone who invents an new, original and ornamental design
2. **Plant Patents** – To anyone who invents distinct and new variety of plant



Petents in Computer Security : RSA Public-Key, granted 1983 and expired in 2000. Patent holder receive a fee for each RSA implementation.

Intellectual Property – Trademarks

Trademarks

A trademark is a word, name, symbol, or device that is used in trade with goods. Indicate the source of the goods and to distinguish them from the goods of others.

✓ Prevent others from using a confusingly similar marks

✗ Not to prevent others from making the same goods, or from selling it





Intellectual Property – Relevant

Intellectual Property – relevant to Network and Computer Security

1. **Software –** Operating system, utility programs, applications, shareware. Copyright protection is available, in some cases, Patent protection also be appropriated.
2. **Databases –** e.g economics forecasting db, protected by Copyright
3. **Digital Content-** Audio files, video, multimedia, courseware, website content, digital devices, protected by copyright
4. **Algorithm** – RSA Public Key, protected by Patent

Privacy

What is Data Privacy?

"Also known as Information Privacy – data protection that addresses the proper storage, access, retention, immutability and security of sesentive data"

Personal Data : *names, addresses, ic number, student id, staff id*

Other valuable data : *financial data, IP, personal health data*



Data privacy can be protected by :

- Data Backups
- Disaster Recovery Considerations / Plan (DRP)
- Data Security

Goals: To ensure privacy and security of sensitive data, maintaining availability & consistency

Data Privacy

Data Privacy – to keep data safe against improper access, theft or loss.

- Keep data confidential and secure by proper data management.
- Prevent unauthorized access resulting in data loss, alteration and theft.



Data Breaches – serious effect on all parties involved.

1. **Individual –** Improper financial and credit activity, use of IC No to register Astro, unifi etc. Compromised social media accounts.
2. **Business –** Regulatory consequences, fines, lawsuits, brand damage, bad reputation.

Data Privacy vs Data Security

Data Privacy – a subset of data security.

- Data privacy can't exist without data security.

There are closely related ideas:

1. **Data Privacy** – Focuses on issues related to collecting, storing, and retaining and transfer data, under laws such as GDPR, HIPAA.
2. **Data Security** – Protection of data against unauthorized access, loss or corruption, via data encryption, hashing and tokenization to guard data.



Ethical Issues

There are many potential misuses and abuses of information and electronic communication that create privacy and security problems

Issues facing Cybersecurity Professionals?



✓ CONFIDENTIALITY – They handle personal, private or proprietary Information that should be kept strictly confidential.

✓ SECURITY - Responsible for following appropriate cybersecurity procedures in their lives, multiply by 100 times more strict.



Ethical Issues

Ethical issues that arises from computers technology and information systems:

1. Computer is the primary repository of personal information, bank records and other financial information.
2. Other types of database with considerable value

- ✓ These assets can be viewed, created, altered by technical and automated means.
- ✓ People that have the power related to those assests can exploit the technology and access permission

Ethical Hacking & Penetration Testing

What is Hacking?

Breaking into computer and network system to steal information or cause a damage to the target by several ways like intruding, attacking and blocking

The Hacker :

A hacker is someone who seeks and exploits weaknesses in a computer system and network



Ethical Hacking & Penetration Testing

Types : Hacking vs Ethical Hacking

There is a different between both term.



The Difference is : ***PERMISSION***

- ✓ Hacking without permission is illegal and each country has their own act to govern this activities.
- ✓ Ethical hacking is hacking or security testing activites with written consent from the client to detect potential security threats.
- ✓ Normally conducted by trusted security professional from internal or external organization.

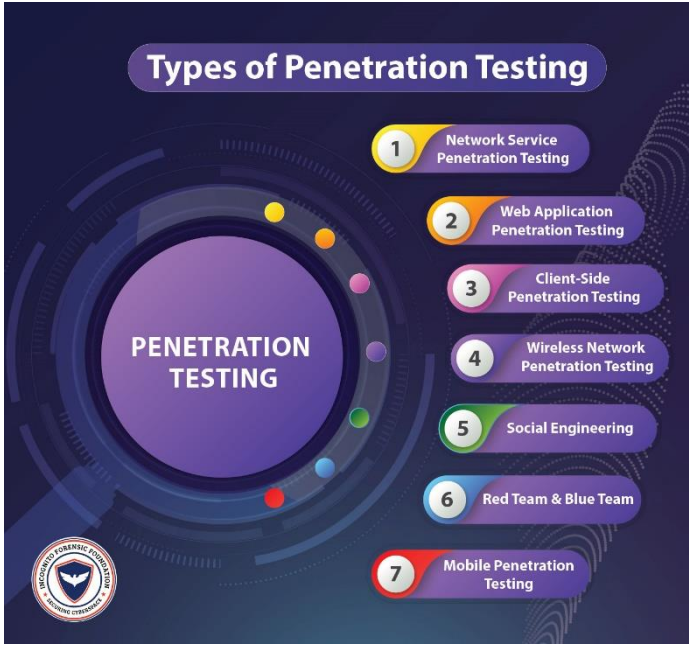
Ethical Hacking & Penetration Testing

What is Penetration Testing?

Penetration Testing is the process to find vulnerability, loopholes, weaknesses in a system and network.

- ✓ Use the same attacking methods and tools like the real attacker
- ✓ Conducted by trusted individual (Internal / External) with Formal Authorization / Permission
- ✓ Will produce report and recommendation of Pen Test findings

Also Known as : Pen Testing, Security Assessment, Ethical Hacking, Pen Test, White Hat Hacker



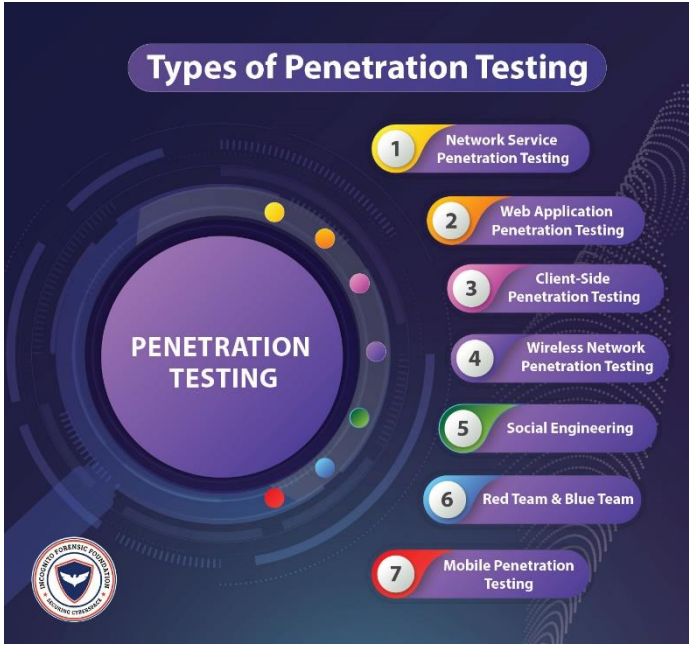
Icognito Forensic Foundation

Ethical Hacking & Penetration Testing

The Importance of Penetration Testing

1. To find vulnerabilities in your network, systems and applications before the attacker do
2. To confirm with the effectiveness of the current security measure in place
3. To test the security response team and network admin
4. To test IPS, IDS, WAF and other security devices
5. To test a new services or system before being online/ deploy
6. To comply with some policy or guideline like HIPAA and PCI-DSS

To see what happen after the attack, what data the attacker can gain etc



Icognito Forensic Foundation

Ethical Hacking & Penetration Testing

Pen Test Methodology by EC-Council Global Services (EGS)

The EGS Methodology

- **Information Gathering:** Collect as much information as possible to gain a better understanding of the test environment.
- **Scanning and Evaluation:** Perform an automated scan on the target along with manual verification of findings.
- **Exploitations:** Exploit the vulnerabilities identified from the scanning phase through the use of both automated and manual techniques
- **Verification:** Conduct manual verification and analysis to validate all the findings based on test cases and standards.
- **Report:** Document all verified findings with their severity rating base on best practices and standard rating scores.



Computer Security And Law

US and General

1. Sarbanes-Oxley (SOX)
2. Computer Security Act of 1987
3. Health Insurance Portability and Accountability Act (HPAA) - Ensures patient confidentiality for all healthcare-related data.



Computer Security And Law (Malaysia)

The Copyright Act 1997

- Provides a very sufficient protection of copyrightable work such as computer software

Computer Crime Act 1997

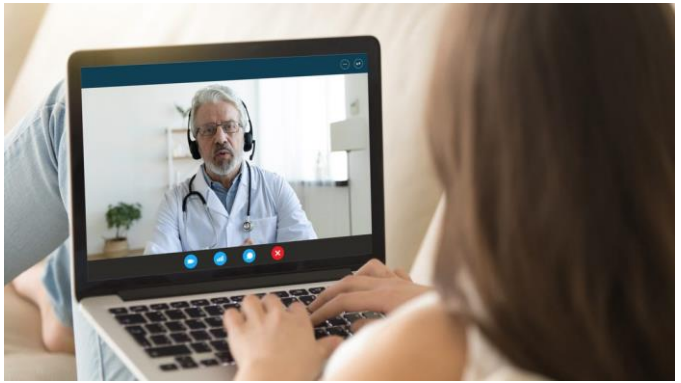
- Impose penalties on fraudulent, dishonest acts and misuse of computer which considered as a crime.
- E.g Unauthorized access, U.A for further offenses and unauthorized modification.



Computer Security And Law (Malaysia)

Telemedicine Act 1997

- An act to prewire for the regulation and control of the practices or the application of telemedicine in Malaysia and anything related to it



Digital Signature Act 1997

- Enforced in 1998.An act legalizes the legal recognition and authentication of the cretor of an electronic document

Computer Security And Law (Malaysia)

Communication and multimedia Act 1998

- Enforce in 1999
- Policy, Telecomm, Broadcasting

Malaysia Communication and Multimedia Commission act 1998

- Supervise and regulate the comm and multimedia activities in Malaysia based in above law



Computer Security And Law (Malaysia)

Optical Disk Act 2000

- License to manufacture optical discs
- The Act aims to reduce piracy of IP by require all makers of optical disc to obtain license from MITI

Electronic Commerce Act 2006

- Legal recognition of electronic messages in commercial transactions
- The use of the electronic messages to fulfil legal requirements
- Facilitate commercial transactions through the use of electronic means



SECURITY RESOURCES

1. CERT (www.cert.org)
2. Microsoft Security Advisor (www.microsoft.com/security/default.mspx)
3. F-Secure Corporation (www.f-secure.com)
4. SANS Institute (www.sans.org)
5. Security Focus
6. CyberSecurity Malaysia
7. https://www.techtarget.com/searchcio/definition/data-privacy-information-privacy#:~:text=Data%20privacy%20is%20a%20discipline,data%20loss%2C%20alteration%20or%20theft.



Security Resources

1. https://egs.eccouncil.org/remote-penetration-testing/
2. https://ifflab.org/types-of-penetration-testing-by-cybersecurity-professionals/
3.





Summary

- Cybercrime and Computer Crime
- Intellectual Property – Copyright, Patent, Trademark
- Privacy – Data Privacy, Data Security
- Ethical Issues
- Ethical Hacking & Penetration Testing – Overview, Methodology
- Legal Aspect – International Law, Malaysian Law